Cybersecurity Portfolio: File Permissions in Linux

Role: Information Security Analyst

Environment: Linux (Bash Shell, Debian-based)

Target Directory: /home/analyst/projects

1. Project Description

As part of the security operations team supporting the research department, I was tasked with auditing and restructuring file and directory authorizations within the projects directory. The initial access rights did not adhere to the principle of least privilege, posing potential security risks to research assets. My objective was to evaluate the current access configuration via Linux command-line tools, identify policy deviations, and apply granular permission controls using chmod to ensure data confidentiality and integrity.

2. Check File and Directory Details

To conduct an initial security audit of the directory structure and uncover any hidden configurations or files, I executed the following command:

```
ls -la
```

Command Output:

```
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-rw--w---- 1 researcher2 research_team   46 Dec  2 15:27 .project_x.txt
```

```
drwx--x--- 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_t.txt
```

Analysis:

- ls: Lists directory contents.
- -l: Enables long-listing format, providing details such as file types, permissions, ownership (researcher2), group associations (research_team), sizes, and timestamps.
- -a: Displays hidden entries starting with a dot (.), uncovering the hidden file .project_x.txt alongside standard project files and the drafts directory.

3. Describe the Permissions String

In Linux systems, the initial 10-character field represents the resource type and permission matrix:

- **Position 1 (File Type):** A hyphen (-) specifies a regular file, whereas a d indicates a directory.
- **Positions 2–4 (Owner / User):** Define read (r), write (w), and execute (x) rights for the resource owner.
- **Positions 5–7 (Group):** Define permissions granted to members of the assigned group.
- **Positions 8–10 (Others):** Define permissions applicable to all other system users outside the user and group categories.
- A hyphen (-) in any permission slot indicates that the specific right is revoked/denied.

Practical Example: project_t.txt (-rw-rw-r--)

- **Leading -:** Confirms it is a regular data file.
- **rw- (Owner):** Read and write permissions enabled; execution disabled.
- **rw- (Group):** Read and write permissions enabled; execution disabled.
- **r-- (Others):** Restricted strictly to read-only access.

4. Change File Permissions

Organizational compliance requires that external/other users must never have write access to internal project files. An inspection revealed that project_k.txt had overly permissive permissions (-rw-rw-rw-), allowing others to modify the file.

Execution:

```
chmod o-w project_k.txt
```

Verification:

```
ls -la
```

```
-rw-rw-r-- 1 researcher2 research_team 46 Dec  2 15:27 project_k.txt
```

Outcome: Write permissions were revoked from other, ensuring that only the designated owner and group retain modification privileges.

5. Change File Permissions on a Hidden File

The research team archived .project_x.txt and specified that no entity should possess write rights to prevent accidental tampering, while both the owner and the group require read access. The file initially had -rw--w----.

Execution:

```
chmod u-w,g-w,g+r .project_x.txt
```

Verification:

```
ls -la
```

```
-r--r----- 1 researcher2 research_team 46 Dec 20 15:36 .project_x.txt
```

Outcome: The owner lost write access, the group had write revoked and read granted. Now, only researcher2 and research_team can view the archived content without any capability to edit or delete its data.

6. Change Directory Permissions

To enforce strict project privacy, management required that only researcher2 retain access to the drafts subdirectory, completely barring group members from entering or inspecting drafts. Because directory access requires execute (x) permission to traverse into the folder, I removed the group's execute rights.

Execution:

```
chmod g-x drafts
```

Verification:

```
ls -la
```

```
drwx------ 2 researcher2 research_team 4096 Dec  2 15:27 drafts
```

Outcome: The permissions changed from drwx--x--- to drwx------, isolating the directory exclusively to the owner and preventing unauthorized group navigation.

7. Summary

Through systematic auditing with ls -la and targeted permission modifications using chmod, I successfully brought the projects directory into full compliance with internal security guidelines. Unauthorized write privileges were revoked from public users on active files, archived records were rendered tamper-proof via read-only enforcement, and sensitive draft directories were restricted solely to the primary researcher. This project reinforces core access control hygiene and practical endpoint hardening within a Linux environment.